

MINDANAO STATE UNIVERSITY – ILIGAN INSTITUTE OF TECHNOLOGY

College of Computer Studies

LABORATORY ACTIVITY 1

REAL-TIME MULTISENSOR ROOM MONITORING SYSTEM

Course: BCA182 – Embedded Systems Programming

Student: Charles Daniel B. Estrada

Platform: STM32 Blue Pill (STM32F103C8) + FreeRTOS

Simulation: Wokwi

Language: C/C++ with STM32 HAL and FreeRTOS

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1. Problem and Requirements

The laboratory problem is to implement a real-time multisensor room monitoring system using an STM32 Blue Pill and FreeRTOS. The system must monitor temperature, humidity, relative light level, and motion; display a selected measurement on an SSD1306 OLED; support rotary-encoder navigation; provide a temperature alarm; and implement ACTIVE/INACTIVE behavior based on PIR motion.

The final implementation follows the required STM32, Wokwi, and FreeRTOS configuration while retaining the required laboratory task architecture and FreeRTOS concepts.

Requirement	Required behavior	Implementation
FR-01	Temperature monitoring	SensorTask acquires DHT22 temperature.
FR-02	Humidity monitoring	SensorTask acquires DHT22 humidity.
FR-03	Relative light monitoring	SensorTask reads LDR through ADC and reports relative light.
FR-04	Motion detection	MotionTask monitors PIR.
FR-05	OLED selected measurement	DisplayTask renders the selected measurement.
FR-06	Encoder navigation	InputTask changes Temperature/Humidity/Light/Motion modes.
FR-07	Temperature alarm	Alarm logic uses $<18^{\circ}\text{C}$ or $>30^{\circ}\text{C}$.
FR-08	ACTIVE/INACTIVE	StateTask manages the two system states.
FR-09	15-second inactivity	StateTask enters INACTIVE after the timeout.
FR-10	PIR reactivation	Motion notification returns the system to ACTIVE.

2. System Architecture and Design

2.1 Hardware Architecture

The system uses the STM32F103C8 Blue Pill as the controller. The peripherals are DHT22, LDR, PIR, rotary encoder, SSD1306 OLED, buzzer, UART, and the onboard LED. The final Wokwi circuit follows the working configuration pin arrangement.

Component	Pin	Purpose
DHT22	PA0	Temperature and humidity
LDR	PA1 / ADC1 CH1	Relative light level
PIR	PA2	Motion detection
Encoder CLK	PA3	Rotary encoder
Encoder DT	PA4	Rotary encoder
Buzzer	PA8 / TIM1 PWM	Temperature alarm
UART TX	PA9	Serial output
UART RX	PA10	UART pin
OLED SCL	PB6	I2C1 clock
OLED SDA	PB7	I2C1 data
Onboard LED	PC13	Status/task activity

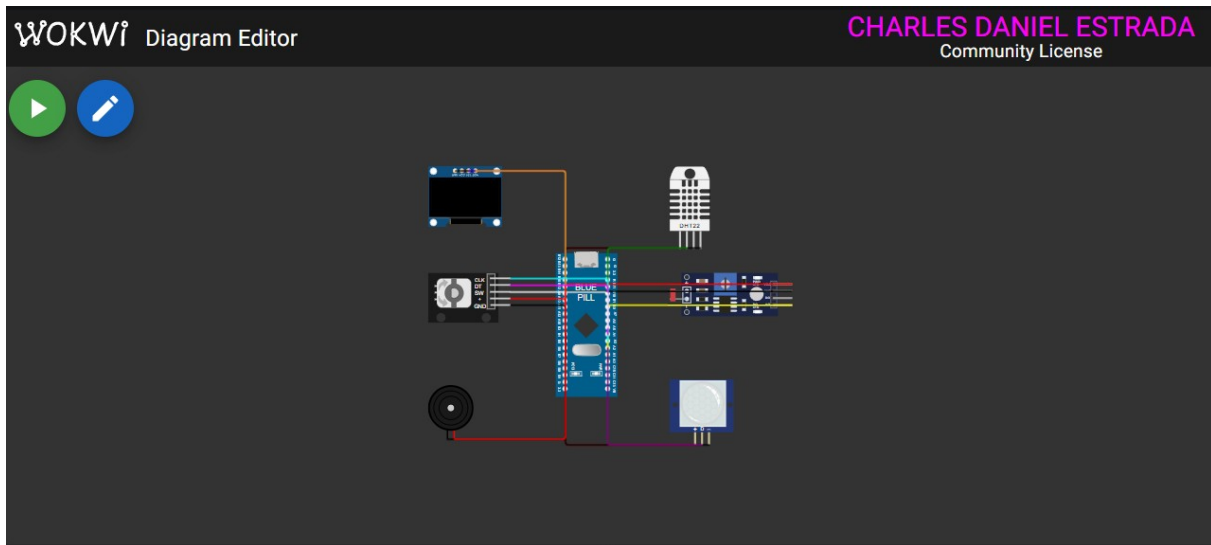


Figure 1. Final Wokwi circuit used for the project.

2.2 Software Architecture

The software is divided into FreeRTOS tasks and peripheral modules. SensorTask produces sensor data, DisplayTask owns the OLED, InputTask handles encoder navigation, MotionTask monitors PIR, AlarmTask handles temperature alarm behavior, and StateTask manages system state. TaskA and TaskB are retained as periodic tasks in the laboratory implementation.

2.3 Subsystem Decomposition

- Sensor subsystem: DHT22 and LDR acquisition with SensorData passed through a queue.
- Display subsystem: SSD1306 initialization and rendering owned by DisplayTask.
- Input subsystem: rotary encoder mode selection.
- Motion/state subsystem: PIR input, MotionTask notification, and StateTask state transitions.
- Alarm subsystem: temperature evaluation and TIM1 PWM buzzer control.
- Communication/synchronization subsystem: queue, mutexes, task notification, and critical sections.

2.4 State Machine

The system starts in ACTIVE. If no motion is detected for 15 seconds, StateTask transitions to INACTIVE. In INACTIVE, unnecessary display and alarm activity is reduced while motion detection remains available. A PIR event returns the system to ACTIVE.

3. FreeRTOS Architecture

FreeRTOS is used to separate sensor acquisition, display handling, user input, motion monitoring, alarm processing, and system-state management. Tasks use explicit priorities and blocking behavior rather than continuous busy loops where periodic scheduling is required.

3.1 Required FreeRTOS Task Table

Task	Responsibility	Trigger / Period	Priority	IPC	Typical Blocked Condition
SensorTask	Measurements	2 s	2	Queue	vTaskDelayUntil()
DisplayTask	OLED update	200 ms active / 500 ms inactive	2	Queue / mutex	vTaskDelayUntil()
InputTask	Encoder input	1 ms periodic	3	Mutex	vTaskDelay()
MotionTask	PIR monitoring	100 ms periodic	3	Task notification	vTaskDelay()
AlarmTask	Alarm logic	500 ms periodic	2	Queue	vTaskDelay()

StateTask	ACTIVE/ INACTIVE state	Notification / 15 s timeout	2	Task notification	Notification wait / timeout
TaskA	Periodic task	1 s	2	Serial mutex	vTaskDelay()
TaskB	Periodic task	1.5 s	1	Serial mutex	vTaskDelay()

3.2 Technical Justification

SensorTask uses a two-second periodic delay because environmental measurements do not require continuous sampling. DisplayTask uses vTaskDelayUntil() so display refresh remains periodic without a continuously running loop. MotionTask and InputTask have higher priorities because their inputs are interactive. StateTask blocks while waiting for a motion notification or until the inactivity timeout. The sensor queue avoids unsynchronized global-data access, while the serial mutex prevents concurrent UART transmissions from interfering with one another.

4. Implementation

The implementation uses the required STM32, FreeRTOS, and Wokwi configuration. The laboratory architecture, task names, priorities, synchronization mechanisms, state machine, and unit tests were preserved.

4.1 FreeRTOS and Startup

The project uses the Wokwi-compatible ARM Cortex-M3 FreeRTOS port. The port uses SVC-based context switching and TICKINT-based critical-section handling. The final clock configuration follows the configured implementation at 8 MHz HSI. The OLED initialization is performed by DisplayTask.

4.2 Sensor Implementation

DHT22 acquisition uses DWT cycle-counter timing, direct GPIO access, and a protected 40-bit read. The LDR uses ADC1 channel 1 on PA1 and reports a relative light value rather than calibrated lux.

4.3 Display and Input

The SSD1306 OLED uses the reference initialization sequence and horizontal addressing configuration. DisplayTask owns the OLED resource. The rotary encoder uses PA3 and PA4 and changes the selected measurement mode.

4.4 Motion and Alarm

The PIR uses PA2. MotionTask sends a task notification to StateTask. The buzzer uses TIM1 PWM on PA8 with a 1 kHz, 50% duty-cycle configuration when the temperature alarm condition is active.

5. Verification and Testing

5.1 Build Verification

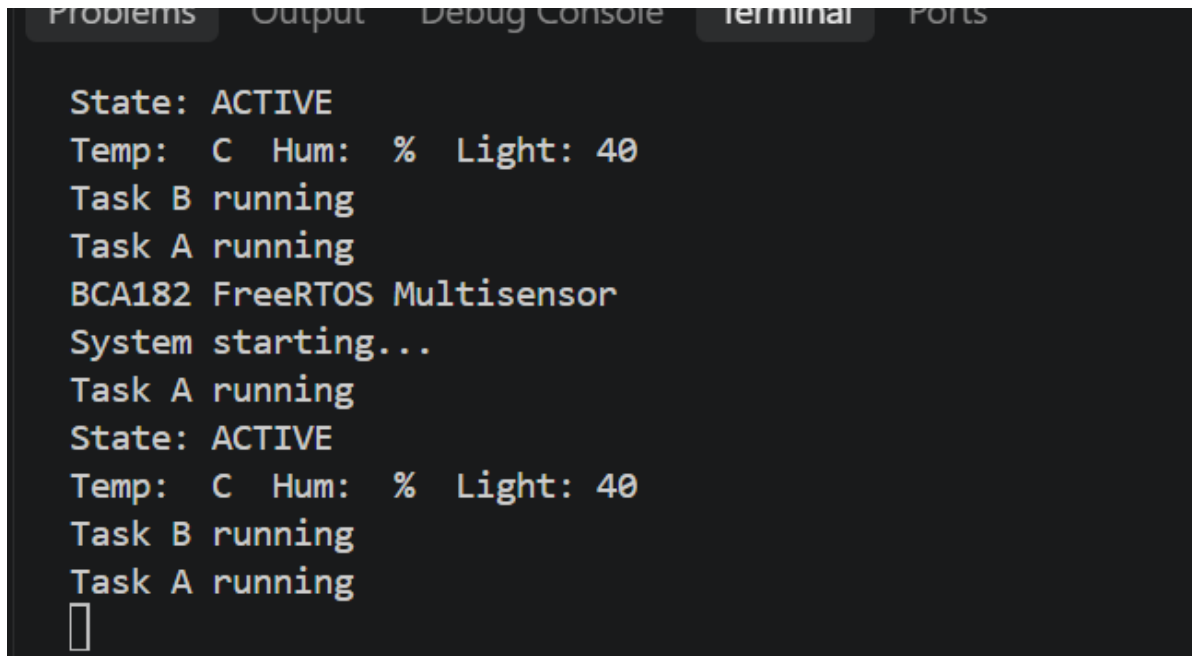
The final STM32 firmware build completed successfully after the configured implementation was adapted. Reported resource usage was 16,116 / 20,480 bytes of RAM (78.7%) and 19,656 / 65,536 bytes of Flash (30.0%).

5.2 Unit Testing

Test Suite	Tests	Result
test_temperature	5	5/5 PASS
test_display_mode	4	4/4 PASS
test_system_state	4	4/4 PASS
Total	13	13/13 PASS

5.3 Wokwi Functional Verification

The project was launched in Wokwi and actual runtime output was observed. The serial output showed the system starting, Task A running, Task B running, State: ACTIVE, and a light-level reading of 40. This confirms that the FreeRTOS runtime is executing in the simulation.



```
Problems Output Debug Console Terminal Ports

State: ACTIVE
Temp: C Hum: % Light: 40
Task B running
Task A running
BCA182 FreeRTOS Multisensor
System starting...
Task A running
State: ACTIVE
Temp: C Hum: % Light: 40
Task B running
Task A running
█
```

Figure 5. Observed Wokwi runtime output.

The available evidence confirms runtime execution, but individual functional tests should only be marked PASS after their specific stimulus and response are observed.

Test	Expected verification	Current evidence
FT-01	Temperature measurement	Not individually verified
FT-02	Humidity measurement	Not individually verified
FT-03	Light-level response	Light: 40 observed; stimulus-response test not fully recorded
FT-04	Encoder clockwise navigation	Not individually verified
FT-05	Encoder counter-clockwise navigation	Not individually verified
FT-06	Alarm above 30 °C	Not individually verified
FT-07	Alarm clears in normal range	Not individually verified
FT-08	PIR produces ACTIVE behavior	ACTIVE state observed; stimulus-response test not fully recorded
FT-09	15 s inactivity produces INACTIVE	Not individually verified
FT-10	PIR reactivates system	Not individually verified

5.4 Deliberate Fault Experiments

Experiment	Controlled fault	Engineering observation
1	Removed a blocking delay	Expected to increase continuous execution and reduce opportunities for other work; restored.
2	Raised a task priority	Expected to affect lower-priority scheduling; restored.
3	Removed serial mutex	Expected to allow concurrent UART output/interleaving; restored.

Where a runtime effect was not consistently observable, the result is described as an expected engineering consequence rather than an observed failure.

6. Static Code Analysis

The final project was checked through the PlatformIO build and native test workflow. The STM32 build completed successfully and the native suite passed 13/13 tests. The patched-port symbols were also verified in the firmware, including SVC_Handler, SysTick_Handler, vPortYield, and vPortGateTick.

The project uses the PlatformIO FreeRTOS dependency for the kernel and a separate patched-port library for the Wokwi-compatible ARM Cortex-M3 port. The old vendored FreeRTOS directory was removed from the repository to prevent a second kernel tree from being compiled.

The static-analysis section should report the exact pio check findings if a final pio check output is available. No unverified severity or warning count is invented in this report.

7. Engineering Discussion

7.1 Wokwi Adaptation and Debugging

The original implementation differed from the known-working configuration in several low-level areas, including FreeRTOS port behavior, DHT22 timing, OLED initialization, clock setup, peripheral pin assignments, and buzzer implementation. The project was configured consistently across the STM32 peripherals, FreeRTOS port, and Wokwi circuit to provide a stable implementation.

7.2 Trade-offs

Using a consistent low-level configuration reduces uncertainty in the Wokwi environment. The final design keeps the laboratory-required task architecture and IPC mechanisms while aligning the peripheral configuration across the system.

7.3 Limitations

- The LDR reports relative light level rather than calibrated lux.
 - The current Wokwi evidence confirms FreeRTOS runtime execution, but not every FT-01 to FT-10 behavior.
 - DHT22 temperature and humidity values have not been individually verified in the presented runtime evidence.
 - Physical hardware may differ from Wokwi because of wiring, electrical characteristics, and timing.
- In the current Wokwi setup, after changing the DHT22 temperature value, the simulation needs to be restarted for the new temperature setting to be reflected correctly in the running system.

7.4 Alternative Approaches

The implementation prioritizes a consistent FreeRTOS port, peripheral timing, OLED initialization, and Wokwi circuit configuration while preserving the required application architecture.

8. Conclusion

The Real-Time Multisensor Room Monitoring System was implemented using STM32, FreeRTOS, and STM32 HAL. The project builds successfully, passes 13/13 native unit tests, and has been observed running in Wokwi with FreeRTOS tasks executing. The implementation demonstrates task scheduling, inter-task communication, synchronization, periodic execution, sensor processing, display management, and an ACTIVE/INACTIVE state machine.

The main improvement made during development was aligning the low-level STM32, FreeRTOS, and Wokwi configuration while preserving the laboratory's required architecture. Further work is to complete and document the individual FT-01 to FT-10 functional observations and any final static-analysis findings before submission.

Requirements Traceability Matrix

This matrix connects each major functional requirement to its implementation and verification evidence, as required by the laboratory instructions.

Requirement	Implementation	Verification
FR-01	SensorTask / DHT22	FT-01; individual runtime observation required
FR-02	SensorTask / DHT22	FT-02; individual runtime observation required
FR-03	SensorTask / LDR	FT-03; light reading 40 observed
FR-04	MotionTask / PIR	FT-08; individual stimulus-response observation required
FR-05	DisplayTask / OLED	FT-01–FT-03; individual display observation required
FR-06	InputTask / encoder	Unit display-mode tests + FT-04/FT-05
FR-07	AlarmTask / temperature logic	Temperature unit tests + FT-06/FT-07
FR-08	StateTask / state machine	State unit tests + runtime ACTIVE observation
FR-09	StateTask / 15 s timeout	FT-09; individual observation required
FR-10	MotionTask → StateTask notification	FT-10; individual observation required